

Outpatient Patient Scheduling & Care System

Product and Functional Requirements

Owner	ABC Health Care Company
Document type	Product Requirements Document (PRD) with Functional Requirements
Version	1.0
Date	December 20, 2025

This document defines the product scope and functional requirements for ABC Health Care Company’s outpatient patient scheduling and care workflow system. It is designed to be implementation-ready and compatible with integration to an existing EHR/EMR.

Intended audience: Product, Engineering, Clinic Operations, Clinical Leadership, IT/Security, Compliance

1. Product Overview

System name: Outpatient Patient Scheduling & Care System (OSCS)

Owner: ABC Health Care Company

Problem statement

Outpatient clinics often rely on fragmented scheduling tools and manual care coordination, leading to no-shows, inefficient intake, inconsistent follow-up, and limited real-time visibility into capacity and patient flow.

Goals (measurable)

Metric	Target outcome
No-show rate	Reduce by 20-40% using confirmations, reminders, waitlists, and frictionless rescheduling.
Scheduling effort	Reduce time to book an appointment by ~30% via eligibility rules and guided workflows.
Access to care	Improve time-to-next-available appointment by 15-25% through capacity visibility and waitlist fill.
Follow-through	Increase completion of follow-up tasks (labs, referrals, care plan items) by ~20%.
Governance	Ensure audit-ready logs and role-based access for PHI handling.

Non-goals for MVP

- Inpatient bed management or hospital-wide operations.
- Full revenue cycle management (billing may integrate later).
- Replacing the EHR/EMR; OSCS should integrate with existing clinical systems.

2. Users and Roles

Primary personas

- Patient
- Front desk / Scheduling staff
- Clinician (MD/DO/NP/PA)
- Nurse / Care coordinator
- Clinic manager

- System admin / IT
- Billing staff (optional for MVP)

Access model

The system must implement role-based access control (RBAC) with least-privilege permissions. All PHI-relevant actions must be audit logged for ABC Health Care Company governance and compliance.

3. Core Workflows

Scheduling (happy path)

- Patient requests an appointment (portal or call center).
- System finds eligible slots based on provider, location, visit type, and clinic rules.
- Appointment is booked and confirmation is sent.
- Patient completes pre-visit intake forms and consents.
- Day-of check-in (remote or front desk).
- Visit occurs; outcomes are documented.
- Follow-up tasks and care plan items are created and assigned.
- Post-visit communication and next appointment (if needed).

Exceptions

- No-show or late cancellation handling.
- Provider schedule changes requiring reassignment or rescheduling.
- Overbooking policies based on clinic rules.
- Waitlist slot offers when cancellations occur.
- Multi-appointment sequences (e.g., lab + consult).

4. Feature Scope (Modules)

Owned by ABC Health Care Company

- Patient Access and Portal
- Scheduling and Calendar Management
- Intake and Check-in
- Care Plan and Tasking (Care Management)
- Communication (SMS/Email/Portal Messaging)
- Clinic Operations (Queue, Rooming, Status Board)
- Reporting and Analytics
- Administration and Configuration
- Integrations (EHR/EMR, Messaging, Telehealth, Labs/Referrals)

5. Functional Requirements

A. Patient Portal and Access

FR-001 Patient registration and identity

- Patients can create accounts using email and/or phone.
- Support OTP verification; optional identity checks can be enabled by policy.
- Login attempts are rate-limited; authentication events are audited.

FR-002 Patient profile management

- Patients can manage demographics, contact info, preferred language, and communication preferences.
- Patients can add emergency contact information.

FR-003 Appointment request and self-scheduling

- Support instant booking for eligible visit types and a request-and-approve mode when required.
- Patients can search by visit type, location, provider preference, and time window.
- Only eligible time slots are displayed; double-booking is prevented unless allowed by policy.

FR-004 Reschedule and cancel

- Patients can reschedule/cancel within configured policy windows.
- Cancellation reason is captured (structured list plus optional free text).
- When a slot opens, the waitlist workflow is triggered (if enabled).

FR-005 Forms and documents

- Patients can complete intake forms, questionnaires, and consents digitally.

- Support e-signature for required consents.
- Signed documents are immutable and versioned; completion status is visible to staff.

B. Scheduling and Calendar Management

FR-101 Provider calendars

- Define working hours, schedule templates, breaks, and buffers.
- Template changes support effective dates to avoid breaking published schedules.

FR-102 Multi-location scheduling

- Support multiple clinic locations with distinct hours and resources.
- Support cross-location search for availability.

FR-103 Visit types and duration rules

- Visit types define default duration, required resources, prerequisites, and allowed booking channels.
- Visit types can be configured as staff-only or patient self-scheduling.

FR-104 Resource scheduling

- Appointments can reserve rooms and equipment where required.
- Conflicts are detected across provider, room, equipment, and staff resources.

FR-105 Waitlist and fill logic

- Patients can opt into a waitlist with preferred time windows and acceptable providers/locations.
- When a slot opens, offers are sent in priority order and expire after a configurable time.
- If not accepted, the slot is released back to availability.

FR-106 Overbooking rules

- Overbooking policies are configurable by provider, visit type, and time-of-day.
- When overbooking is used, staff must enter a justification; overbooked visits are visually flagged.
- All overbooking actions are audited.

FR-107 Scheduling constraints and eligibility

- Enforce constraints such as new vs established patient rules, age limits, insurance acceptance, and referral requirements.
- If constraints fail, the system blocks booking or routes to a request-and-review workflow.

FR-108 Appointment lifecycle states

- Appointment states include Requested, Scheduled, Confirmed, Checked-in, In-room, Completed, Cancelled, No-show.
- State transitions are validated and audited.

C. Intake, Check-in, and Clinic Flow

FR-201 Pre-visit intake

- Automated reminders are sent for incomplete intake items.
- Staff can view missing items per patient/appointment.

FR-202 Check-in modes

- Support front desk check-in and mobile self check-in (policy-gated by time and optional geo-fence).
- Optional capture of insurance card and ID images.
- Co-pay collection can be integrated (optional).

FR-203 Clinic status board

- Real-time board shows patient flow: arrived, waiting, roomed, clinician assigned, completed.
- Filters by provider, location, and visit type.

D. Care Management (Post-visit Follow-through)

FR-301 Care plan creation

- Clinicians and care team members can create care plans tied to the encounter.
- Care plans include goals, interventions, tasks, and due dates.

FR-302 Task management

- Tasks can be assigned to patients or staff; tasks include owner, due date, priority, and status.
- Overdue tasks escalate based on policy; reminders are sent as configured.

FR-303 Follow-up scheduling from care plan

- Generate follow-up appointment suggestions based on care plan tasks.
- Provide a 'schedule next' workflow from visit documentation.

FR-304 Care team messaging

- Internal messaging supports @mentions and templates (optional in MVP).
- Patient messaging is handled separately via the communication module.

E. Communication and Notifications

FR-401 Notification engine

- Support SMS and email notifications (MVP); optional IVR/push in later phases.
- Templates include confirmation, reminders (configurable timing), intake pending, cancellation/reschedule, waitlist offer, and post-visit instructions.
- Opt-in/opt-out preferences are stored and enforced; delivery status is tracked.

FR-402 Two-way messaging

- Patients can reply to SMS or message in the portal.
- Keyword routing supports automated actions (e.g., CANCEL) and staff inbox routing.
- Messages containing PHI are stored securely and audited.

F. Reporting and Analytics

FR-501 Operational dashboards

- No-show rate, late cancel rate, utilization, lead time, fill rate, and waitlist conversion.
- Staff productivity metrics (appointments booked, requests handled) where applicable.

FR-502 Clinical follow-through

- Task completion rates, follow-up adherence, referral completion.
- Clinic and provider-level views with role restrictions.

FR-503 Export and BI

- CSV export with role-based restrictions and audit logging.
- Read-only API access for BI tooling (optional).

G. Administration and Configuration

FR-601 Clinic setup

- Manage locations, hours, holidays, rooms, and equipment.
- Manage providers, roles, specialties, visit types, intake requirements, and policy rules.

FR-602 Templates

- Appointment templates by provider/location/visit type.
- Form templates with versioning and effective dates.

FR-603 Audit logs

- Track create/update/delete for appointments, patient demographics, care plans/tasks, and permissions.
- Audit entries include actor, timestamp, and before/after values; justification is captured when required.

H. Integrations

FR-701 EHR/EMR integration

- Support HL7 v2 and/or FHIR for patient and appointment data exchange.
- Define source-of-truth and conflict resolution strategy.

FR-702 Messaging provider integration

- Integrate with an SMS gateway and email provider; store message metadata and delivery receipts.

FR-703 Telehealth

- For tele-visit types, generate and attach a meeting link at booking (optional).

FR-704 Insurance eligibility

- Eligibility checks prior to confirmation (optional; depends on payer integration).

6. Data Requirements (High-level Entities)

All data is managed under ABC Health Care Company governance and access policies. Key entities include:

- Patient
- Provider (Practitioner)
- Location
- Resource (Room/Equipment)
- Appointment
- Visit Type
- Schedule Template / Availability Block
- Encounter (Visit Summary)
- Intake Form + Response
- Consent Document + Signature
- Care Plan
- Task
- Message/Notification
- Audit Log
- User + Role + Permission

7. Non-functional Requirements

Security and privacy

- Role-based access control (RBAC) with least privilege and separation of duties where appropriate.
- Encryption in transit (TLS 1.2+) and at rest (industry-standard encryption).
- Immutable audit logs for PHI-relevant actions.
- Optional MFA for staff; configurable session timeouts.
- Rate limiting and basic anomaly detection for authentication and portal usage.

Reliability and availability

- Target availability: 99.9% (configurable by ABC Health Care Company).
- Automated encrypted backups with defined RPO/RTO targets.
- Integration failures handled via retry queues and operational alerts.

Performance

- Availability search returns results in under 2 seconds for typical clinic data volumes.

- Clinic status board updates within 5 seconds under normal load.

Compliance and auditability

- Audit logs retained per ABC Health Care Company policy; exports restricted by role.
- Support patient access requests and record export processes per organizational requirements.

8. MVP Recommendation

MVP scope for ABC Health Care Company

- Staff scheduling with provider calendars, visit types, booking, and reschedule/cancel.
- Patient portal self-scheduling for limited visit types plus confirmations and reminders.
- Basic waitlist capability.
- Intake forms and check-in workflows.
- Basic care tasks and follow-up tracking.
- Operational dashboards (no-show and utilization) and audit logs.
- Baseline EHR integration for Patient and Appointment synchronization.

Phase 2 (post-MVP)

- Advanced resource constraints and capacity optimization.
- Two-way messaging with smart routing and ticketing integration.
- Insurance eligibility checks and payment/co-pay collection integrations.
- Telehealth integration expansion and deeper clinical documentation workflows.
- Advanced analytics and forecasting.

9. Definition of Done

- All appointment state transitions are validated and audit logged.
- Role permissions verified with a test matrix covering all personas.
- Security baseline tests (SAST and basic penetration testing) completed.
- Load testing validates scheduling search and status board performance targets.
- Integration failure handling validated (queue, retry, alert).
- UAT scripts for top workflows executed and signed off by ABC Health Care Company stakeholders.